

AIR TRANSPARENT SOUNDPROOF WINDOW

Abstract

This study presents the design, optimization, numerical modelling, and experimental validation of air-transparent acoustic metamaterial windows aimed at achieving simultaneous noise reduction and natural ventilation. The numerical model was initially validated analytically using classical muffler theory and existing literature. Various resonant metasurface-inspired unit-cell configurations were then analysed using ANSYS. The investigation began with three square-cavity configurations (90 mm, 45 mm, and 22.5 mm), where predicted resonant frequencies closely matched the transmission loss (TL) peaks, confirming resonance-driven attenuation. Among these, the 22.5 mm cavity demonstrated the best performance, achieving TL values between 25 and 37.5 dB in the 900–1500 Hz range, attributed to enhanced impedance mismatch and damping effects. To optimize the trade-off between acoustic insulation and airflow, 432 design configurations were generated and analysed. Results showed that smaller cavity sizes improved TL, while larger apertures reduced pressure loss and enhanced ventilation. Correlation analysis identified aperture diameter as the most influential parameter for both acoustic and flow performance. A surrogate-assisted multi-objective optimization, employing Random Forest regression and NSGA-II, identified Pareto-optimal solutions, with a selected knee-point design offering improved TL alongside moderate pressure drop. Building on these optimized results, Design B was further refined with hole diameters of 40 mm, 50 mm, and 60 mm. Multi-hole configurations significantly enhanced acoustic attenuation, achieving TL values up to 30–50 dB due to increased scattering and path-difference effects. Finally, a frequency-selective resonant design targeting low-frequency attenuation was experimentally validated via insertion loss measurements. The experimental outcomes closely matched simulations, confirming the methodology's effectiveness.